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Design and Decision-making Processes Behind SDU's Agile Production Cells

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1. Introduction

High mix/low volume describes production scenarios, where manufacturers offer many product variants between which production is often switched [1], hence requiring the production equipment and production control to feature the necessary agility to easily and/or automatically shift and adapt to the next product variant (agile production). Closely linked to the concepts of Industry 4.0, agile production aims for robotizing all sorts of production operations – with a special focus on assembly, where any complexities resulting from part and product varieties today are still mostly countered by commissioning manual operations [2]. Attempts to transfer these manual assembly operations to fully automated robotic assembly cells thus need to systematically address rapidly evolving varieties with lean and reconfigurable approaches that eventually allow for implementing similar levels of agility as with manual operations.

SDU Robotics designed and implemented a novel robotic assembly cell to facilitate, showcase and benchmark this transfer of conventional means of production to robotized, agile production (see **Fig. 1**). The cell successfully competed in the industrial category of the World Robot Challenge (WRC), that the Japanese Ministry of Economy, Trade and Industry sponsored as part of the World Robot Summit taking place in Tokyo, Japan, 17–21 October 2018 [3].

This contribution aims for describing and discussing the design processes leading to SDU' agile production cell in more detail. Section 2 explores how the design processes may be linked and influenced by Danish culture and Danish design thinking. Section 3 gives an overview over the overall design and development processes that led to the cell, which was at designed in the 9 months before the WRC as its first deployment. Based on the comprehensive cell description in [3], Section 4 and Section 5 recount the aspects of the hardware and software design of the cell. Finally, Section 6 closes the contribution with some closing thoughts and remarks.



Fig. 1 SDU's agile production cell

2. Relations to Culture & Design

"Form follows function" is a design maxim originally coined by Greenough [4], and was later interpreted by Sullivan and Wright in the US and the Bauhaus school of design in Europe as a commitment to fully align the design of an artifact with its intended use and not to obstruct the intended use with unnecessary features and/or decorations [5] [6]. Based on Bauhaus' heavy impact on modern industrial design worldwide, the design maxim was further propagated and evolved and even influences today's engineering methods, such as software engineering [7]. However, by focusing on functionality alone, engineers drive "form follows function" to its limits. Norman analyzed everyday object designs and documented how solely function-driven design negatively impacts ease-of-use due to function overload and a lack of clear "affordances" - design hints that guide users to best usage [8]. In this context, Danish design thinking is acclaimed for a good balance between sim-

原稿受付 2020 年 7 月 2 日
キーワード：Robot, Assembly, Agile Production, Design, Agile Development
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plicity and aesthetics while still providing a solid everyday usability [9] [10]. Four key factors of Danish culture and Danish design thinking have played into the design processes described here [11]:

Equality & Availability. Equality of people and equal opportunity of access to resources and decisions is central to Denmark and the other Scandinavian countries, and design is supposed to be widely available, used and visible in public as well as private spaces.

As it characterizes also organizational structures in Denmark such as the organizational structures at SDU Robotics, the notion of equal word and influence on decisions certainly influenced the general design process as well (see Section 3).

Simplicity & Sturdiness. Rooted in Denmark's rough climate and its long history of seafaring and agriculture, Danish culture & design has a preference for sturdy, durable artefacts, with a simple and clear structure and made from long-lasting materials.

The overarching factor of simplicity and sturdiness is certainly visible in the technical design of the cell, which features a clear, robust structure in both, the software architecture (see Section 4) and the hardware design (see Section 5).

It can be debated if another key factor of Danish design, a recognizable aesthetic appeal [11], is applicable for the cell. Some components of the cell, in particular the UR robots, may have been designed by their manufacturers with (Danish) aesthetics in mind. However, as the cell design first and foremost had to support a set of functionalities, the resulting aesthetic expression was mostly circumstantially following from addressing the requirements.

3. General Design and Development Processes

The general design and development processes of the cell were fully initiated in April 2018 and then carried out until the WRC in October 2018. Very much reflecting the organizational structure of SDU Robotics, a consortium of five professors (Full and Associate) jointly decided to apply for participating in the competition and to jointly provide staff, expertise, equipment as well as administrative support to carry out the development. SDU's Industry 4.0 Lab contributed lab space and budgets for the development.

From SDU's perspective, the objectives of the WRC were directly aligned with existing requests from industrial partners to investigate and solve their urgent challenges in factory automation:

- typical assembly operations (pick & place, insertion, screwing)
- handling of rigid object across a range of sizes and weights,
- handling of flexible objects,
- picking objects in uncertain poses,
- picking objects from part bins,
- insertion into movable objects,
- insertion with highly obstructed access,
- reconfiguring for product variants on short notice.

The design and development processes were carried out as an agile R&D activity [12]. Four parallel groups, 1) robot control, 2) computer vision, 3) modelling and simulation, and 4) mechanical design, were working on their specific tasks, and the results were regularly demonstrated resp. presented in order to discuss the progress and to setup the tasks for each group for the next development phase ("sprint"). Each team consisted of 2–5 members.

Quite typical for agile development processes and well-aligned with the organizational structure at SDU Robotics, the decision-making processes in the four groups were made by hearing out recommendations and remarks from all group members, and then (often unanimously) coming to a conclusion on how to proceed. The dynamics of such group decisions were typically not driven by seniority and status, but by hands-on experience with the actual implementation efforts ? the opinions of who are actually facing an issue have the most weight in evaluating potential solutions.

Such design and developments decisions on group level were coordinated by a team leader resp. product owner in the agile development terminology. The role of this team leader for the combined effort was to sit in with the teams and guide and align the efforts across teams to yield a unified result on the cell level. Another aspect of the role was to act as a facilitator and to ensure the best possible working conditions for the groups. The team leader was further supported by i) internal colleagues at SDU, who dealt with administrative tasks such as purchasing processes, ii) external partners at companies, who supported the cell with hardware and software developments.

4. About the Hardware Design

The hardware design of the overall system was very much inspired by earlier cell designs at SDU Robotics [13] [14]. The basic idea was to have a worktable with manipulators and sensors covering the workspace from above. The cell design consists of two mid-size industrial robots on stands (for decoupling the worktable from jerks and vibrations) that are working over a central worktable. The base joints of the two

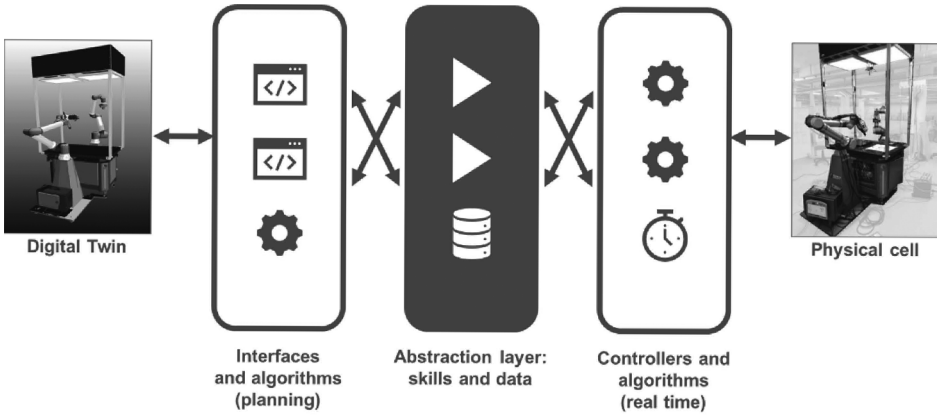


Fig. 2 Software architecture of SDU's cell

robots are rotated by 180 degrees compared to each other (see Fig. 1). The worktable is covered by a sensor frame that allows for carrying light and cameras above to monitor the workspace.

Beyond the industrial assembly challenges posed by WRC, the cell was supposed to be readily and easily deployable for multiple purposes and various processes. The necessary basic hardware for such a level of flexibility and reconfigurability was identified to be i) tabletops for welding [15] and ii) aluminium profiles for technical installations [16]. The tabletops for welding feature a raster of boreholes which can be used to rigidly plug in modular components (self designed or from extensive catalogues). Plus, the tabletops are extremely durable, sturdy and heavy and are thus an ideal support basis for robotic workspaces in factory automation. Similarly, aluminium profiles offer easy (re-)configuration and are extendable from system catalogues as well.

The team at SDU explicitly designed and shaped the overall hardware design of the cell as well as fixtures etc. which were not commercially available or were novel developments. The hardware design of the commercial components was of course decided by the industrial suppliers. The cell consists of two UR10e collaborative robots by Universal Robots (6 DOF / Reach: 1.3 [m]/ Load: 10 [kg]) which are equipped with a built-in force-torque sensor in comparison to previous UR10 models [17]. One robot carries a WSG50 gripper by Weiss Robotics that allows for flexible programming and force control [18]. The other robot carries a 2F85 gripper by Robotiq that offers wide stroke at high forces, flexible programming and force control [19]. Perpendicular to the gripper, the second robot is additionally equipped with an ECSF16 screwdriver by Desoutter that allows for programmable torque/velocity profiles [20]. The sensor frame carries an acA2500-20gc camera by Basler which offers hi-res color images [21]. The FlexCell worktable by Technicon allows for connecting and redistributing power, compressed air and ethernet [22]. For a more detailed view into the cell, see **Fig. 3**.

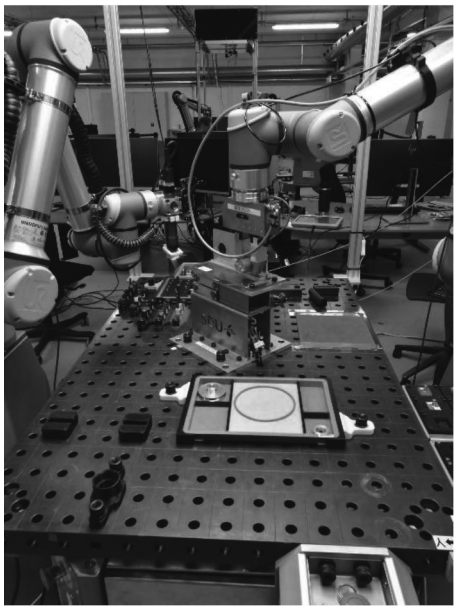


Fig. 3 View into the workspace of SDU's cell

5. About the Software Design

A central focus of the cell design was on the software architecture. The team at SDU Robotics experienced in their previous projects [13] [14] that the eventual usability of industrial robot cell designs is significantly depending how possibilities and complexities arising from the various hardware components are made accessible and manageable by means of a systematic approach to software.

The design of software architecture of SDU's cell was guided by the following design goals resp. features:

Modularity & Reconfigurability. The possibility of exchanging hardware components on demand and to support such reconfigurations by an equivalent exchangeability of driver modules etc. on the software side.

Planning & Simulation. The possibility to plan, sim-

ulate and execute assembly sequences based on digital models of the product and the cell. In addition, to monitor the cell operation based on digital models to allow for data collection and online path planning.

Hierarchy & Distribution. The possibility to define high-level skills (such as "Insertion") based on low-level skills (such as "MovePTP") and to support the distribution of skill across various robot controllers, PLCs and computers in the cell.

The diagram in **Fig. 2** depicts the resulting software architecture of SDU's cell that was designed and implemented to meet the desired features. The basic structure is a service-based architecture [23] – the various components of the physical cell offer low-level skills, e.g. "Open" and "Close" for grippers. These low-level skills are mainly implemented by calling corresponding functionality on the various robot, gripper and sensor controllers. Other such low-level skills are implemented as dedicated algorithms that run under real time conditions, e.g. "SpiralSearch" for insertion execution based on force-torque feedback [3]. The skills are offered as distributed services by means of ROS Action servers [24], which allow for defining and calling remote function calls in ROS networks.

The introduction of skills as services establishes an abstraction layer – all functionality offered by specific physical components and their specific controllers is represented without contacting components and controllers directly, thus decoupling hardware specifics from high-level planning. Further abstraction and decoupling is established by transmitting service parameters (such as target joint poses for "MovePTP") not as direct values, but by fetching them instead from a database by id. This allows for changing execution details without the need to modify the structure of cell operations. In addition, the database supports access control, backup and migration of the poses, paths and parameters that are the central data for cell operation.

The left side of Fig. 2 depicts how planning and simulation are provided for the cell by means of a Digital Twin. The Digital Twin is a comprehensive and virtual replica of the physical cell regarding the exposed functionality as well as the relevant parameters and behaviors. The Digital Twin allows for online robot programming and thereby visualization and verification of robot motions prior to the real-world execution, but also online feedback from the robots and other cell components into the simulation.

6. Conclusion

The design and decision-making processes allowed the team at SDU to develop and implement two instances of the agile production in parallel. The main idea behind this parallel implementation was to make more space and time with the physical cell available for the team – particularly during the run up for the actual WRC competition, where one cell was already sent to Japan, while the other cell was still available in the lab.

The agile development process allowed for scaling and prioritizing tasks based on the overall objectives as well as based on the actual status of the development. The team not only implemented the cells, but also had to ramp up the surrounding infrastructure, including establishing tools and lab space to support such a development. In this regard, the general design and development processes allowed for closely monitoring the progress in the groups and then to down-prioritize specific aspects of the development, such as the Digital Twin. When it became clear that central functionalities for running the WRC challenges were not in place even two months before the competition, the efforts related to the Digital Twin were scaled down and the team was focusing on the more basic functionalities instead.

The resulting performance and winning the WRC was a success for the team [25] and a well-received token of appreciation of the hard work and long hours the team members individually invested in the project. After the WRC, several technologies developed for the cell were migrated and further refined in various other R&D projects at SDU: a gripper finger exchange system, a screwdriver adaption for robotic operation, a new approach to bin-picking based on shoveling, a method for fast setup of pose estimation, a framework for agile programming of robots based on force-torque feedback and programming-by-demonstration [3]. The two implemented cells are used on a daily basis in the context of SDU's Industry 4.0 Lab. Fig. 3 shows the workspace of SDU's cell in detail and gives an impression of the Industry 4.0 Lab environment at SDU today.

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